

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

NAME:JAISURIYA D D

REG NO:192421167

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Technical Presentation (Low)

Assessment Tool Weightage: 35%

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QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.	BL2 – Understand	5
2	Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.	BL2 – Understand	5
3	Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.	BL2 – Understand	5
4	Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.	BL1 – Remember	5
5	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand	5
6	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand	5
7	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand	5
8	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember	5

9	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand	5
10	Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.	BL2 – Understand	5

Code of Conduct:

I JAISURIYA D D (192421167) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

JAISURIYA D D

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

- **Definition of DBMS**
- **Definition**
- **Database Management System (DBMS)** is software that enables users to create, store, retrieve, update, and manage data efficiently in a database.
- **Key Features**
 - Centralized data storage
 - Easy data retrieval
 - Data security
 - Data consistency
 - Multi-user access

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- **Advantages of DBMS**
- **Advantages**
 - Reduces duplicate data.
 - Provides backup and recovery.
 - Ensures secure data access.
 - Improves data sharing.
 - Supports concurrent users.
 - Increases productivity.
 - Makes data management easier.

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Objectives of DBMS

- **Objectives**
- Store data efficiently.
- Reduce data redundancy.
- Ensure data security.
- Improve data integrity.
- Provide fast data access.
- Support multiple users.
- Simplify database management.

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- **What is DBMS?**
- DBMS stands for **Database Management System**.
- It is software used to store, organize, and manage data.
- It allows users to create, update, and retrieve data.
- It helps maintain data in a structured format.
- Examples: **MySQL, Oracle, SQL Server, PostgreSQL**.

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- **Real-World Applications**
- **Applications of DBMS**
- **Banking**
- Customer accounts and transactions
- **Healthcare**
- Patient records and appointments
- **Education**
- Student information systems
- **E-Commerce**
- Product inventory and online orders
- **Airline Reservation**
- Flight booking and passenger details



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2. File-Based System vs DBMS



File-Based System	DBMS
Data stored in separate files	Data stored in a centralized database
High data redundancy	Reduces data redundancy
Low security	High security with user access control
Difficult to share data	Easy data sharing among users
Backup and recovery are difficult	Provides backup and recovery features

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ANSI/SPARC Three-Level Architecture

- **Three Levels:**
- **External Level (View Level):** Shows only the data required by each user.
- **Conceptual Level (Logical Level):** Defines the overall database structure and relationships.
- **Internal Level (Physical Level):** Describes how data is stored on the storage device.

📷 Snapchat

👤 Dharshana

4. Different Data Models

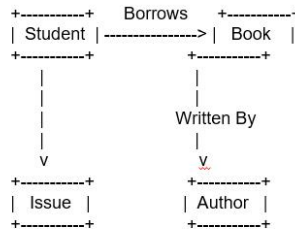
Data Model	Structure	Industry Use Case
Hierarchical Model	Tree structure (Parent → Child)	File systems, Organization hierarchy
Network Model	Many-to-many relationships	Telecommunications, Transportation
Relational Model	Data stored in tables (Rows & Columns)	Banking, Healthcare, E-commerce



5 – ER Diagram for an Online Library System



- **ER Diagram – Online Library System**
- **Entities**
- **Student**
- **Book**
- **Author**
- **Issue**
- **Relationships**
- Student **borrows** Book.
- Author **writes** Book.
- Issue records the **borrow** and **return** details.



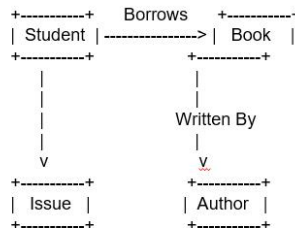
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5 – ER Diagram for an Online Library System



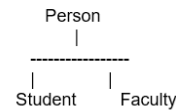
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6 – EER Model: Generalization and Specialization

- EER Model – University Case Study
- What is EER?
- It supports **Generalization** and **Specialization**.
- It represents complex relationships more effectively.
- **Generalization**
- Combines similar entities into a **common parent entity**.
- **Example:** *Student* and *Faculty* → **Person**
- **Specialization**
- Divides a parent entity into **specific child entities**.
- **Example:** **Person** → *Student* and *Faculty*



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7 – Data Independence in DBMS

Data Independence

Definition:

Data Independence is the ability to change the database structure without affecting application programs.

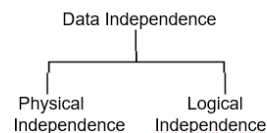
Types of Data Independence

1. Physical Data Independence

- Changes the **physical storage** of data.
- Does **not affect** the logical database structure.

2. Logical Data Independence

- Changes the **logical structure** of the database.
- Does **not affect** user applications.



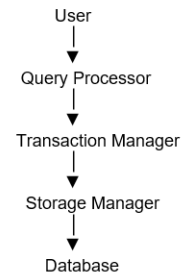
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8 – Software Components of a DBMS

Software Components of DBMS

Main Components

- **Query Processor**
 - Executes SQL queries.
 - Retrieves and updates data efficiently.
- **Storage Manager**
 - Manages data storage on disk.
 - Handles files, indexes, and memory.
- **Transaction Manager**
 - Ensures safe and reliable transactions.
 - Maintains **ACID properties** (Atomicity, Consistency, Isolation, Durability).



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9 – Strong Entity vs Weak Entity

Strong Entity	Weak Entity
Has its own Primary Key	Does not have its own Primary Key
Exists independently	Depends on a Strong Entity
Represented by a single rectangle	Represented by a double rectangle
Example: Student	Example: Dependent

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10 – EER Diagram for Retail Chain Management System

